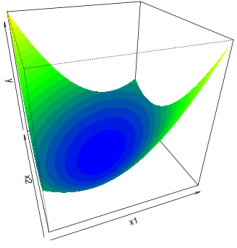
Optimization in Machine Learning

Foundations
Quadratic functions



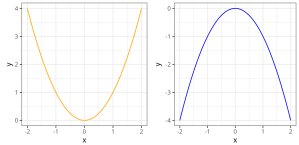
Learning goals

- Definition of quadratic functions
- Gradient, Hessian
- Optima
- Geometry: spectrum and curvature
- Condition number

UNIVARIATE QUADRATIC

- Quadratic function $q : \mathbb{R} \to \mathbb{R}$

$$q(x) = ax^2 + bx + c, \quad a \neq 0$$

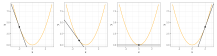


- Left: $q_1(x) = x^2$. Right: $q_2(x) = -x^2$

UNIVARIATE: BASIC PROPERTIES

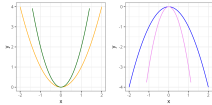
- Slope at $(x, q(x))$:

$$q'(x) = 2ax + b$$



- Curvature:

$$q''(x) = 2a$$



- $a > 0$: q convex, bounded from below, unique global minimum
- $a < 0$: q concave, bounded from above, unique global maximum
- Optimum x^*

$$q'(x) = 0 \;\Leftrightarrow\; 2ax + b = 0 \;\Rightarrow\; x^* = \frac{-b}{2a}$$

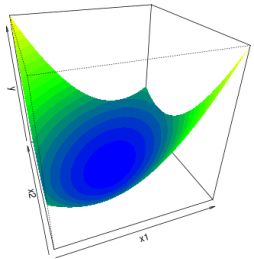
as 2nd derivative: $q''(x^*) = 2a \neq 0$

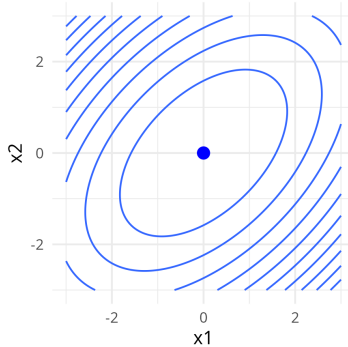
MULTIVARIATE QUADRATIC

- $q : \mathbb{R}^d \to \mathbb{R}$

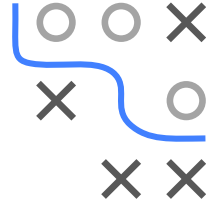
$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} + \mathbf{b}^T \boldsymbol{x} + c$$

- with $\mathbf{A} \in \mathbb{R}^{d \times d}$ full rank, $\mathbf{b} \in \mathbb{R}^d$, $c \in \mathbb{R}$





SYMMETRIZATION



- W.l.o.g. assume $\mathbf{A}$ symmetric, i.e., $\mathbf{A}^T = \mathbf{A}$
- If $\mathbf{A}$ not symmetric, there exists symmetric $\tilde{\mathbf{A}}$ with

$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} = \boldsymbol{x}^T \tilde{\mathbf{A}} \boldsymbol{x} =: \tilde{q}(\boldsymbol{x})$$

- Justification

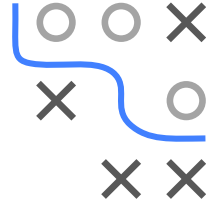
$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} = \tfrac{1}{2} \boldsymbol{x}^T \underbrace{\left(\mathbf{A} + \mathbf{A}^T\right)}_{\tilde{\mathbf{A}}_1} \boldsymbol{x} + \tfrac{1}{2} \boldsymbol{x}^T \underbrace{\left(\mathbf{A} - \mathbf{A}^T\right)}_{\tilde{\mathbf{A}}_2} \boldsymbol{x}$$

- $\tilde{\mathbf{A}}_1$ symmetric, $\tilde{\mathbf{A}}_2$ anti-symmetric ($\tilde{\mathbf{A}}_2^T = -\tilde{\mathbf{A}}_2$)
- Since $\boldsymbol{x}^T \mathbf{A}^T \boldsymbol{x}$ is a scalar, equal to its transpose

$$\begin{aligned} \boldsymbol{x}^T (\mathbf{A} - \mathbf{A}^T) \boldsymbol{x} &= \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} - \boldsymbol{x}^T \mathbf{A}^T \boldsymbol{x} = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} - (\boldsymbol{x}^T \mathbf{A}^T \boldsymbol{x})^T \\ &= \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} - \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} = 0 \end{aligned}$$

- Therefore $\tilde{q}(\boldsymbol{x}) = \boldsymbol{x}^T \tilde{\mathbf{A}} \boldsymbol{x}$ with $\tilde{\mathbf{A}} = \tilde{\mathbf{A}}_1 / 2$

GRADIENT AND HESSIAN



- $q : \mathbb{R}^d \to \mathbb{R}$
$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} + \mathbf{b}^T \boldsymbol{x} + c$$
- Gradient
$$\nabla q(\boldsymbol{x}) = ((\mathbf{A}^T + \mathbf{A})\boldsymbol{x} + \mathbf{b})^T$$
- Under assumed symmetry: $\nabla q(\boldsymbol{x}) = (2\mathbf{A}\boldsymbol{x} + \mathbf{b})^T$
- Directional derivative: $\nabla q(\boldsymbol{x})\, \boldsymbol{v}$
- Hessian
$$\nabla^2 q(\boldsymbol{x}) = (\mathbf{A}^T + \mathbf{A}) = 2\mathbf{A} =: \mathbf{H} \in \mathbb{R}^{d \times d}$$
- Under assumed symmetry: $\mathbf{H} = 2\mathbf{A}$
- Directional curvature: $\boldsymbol{v}^T \mathbf{H} \boldsymbol{v}$

OPTIMUM

- $q : \mathbb{R}^d \to \mathbb{R}$

$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} + \mathbf{b}^T \boldsymbol{x} + c$$

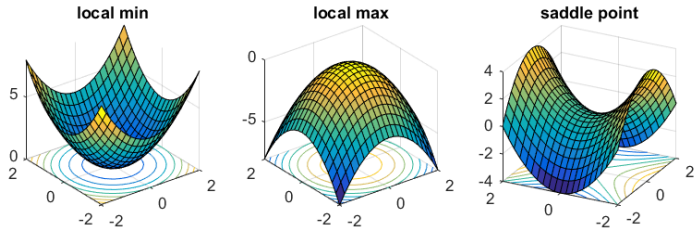
- Since **A** full rank, unique stationary point $\boldsymbol{x}^*$ (min, max, or saddle)

$$\nabla q(\boldsymbol{x}^*) = \mathbf{0}^T$$

$$(2\mathbf{A}\boldsymbol{x}^* + \mathbf{b})^T = \mathbf{0}^T$$

$$\boldsymbol{x}^* = -\tfrac{1}{2}\mathbf{A}^{-1}\mathbf{b}$$

- $q(\boldsymbol{x}^*) = c - \frac{1}{2}\mathbf{b}^T\mathbf{A}^{-1}\mathbf{b}$



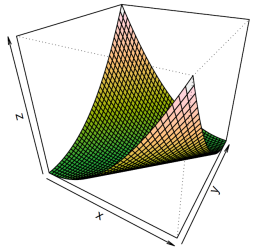
- Left: **A** pos. def. Middle: **A** neg. def. Right: **A** indefinite

OPTIMA: RANK-DEFICIENT CASE

- $q : \mathbb{R}^d \to \mathbb{R}$

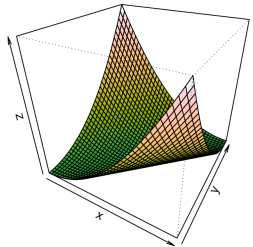
$$q(\boldsymbol{x}) = \boldsymbol{x}^T \mathbf{A} \boldsymbol{x} + \mathbf{b}^T \boldsymbol{x} + c$$

- Assume $\mathbf{A}$ symmetric now
- For stationary points to exist, we need : $\nabla q(\boldsymbol{x}) = 2\mathbf{A}\boldsymbol{x} + \mathbf{b} = 0$
- This implies we need $\mathbf{b} \in range(\mathbf{A})$, let's assume this is the case
- Let $\boldsymbol{x}_p$ be stationary, so $2\mathbf{A}\boldsymbol{x}_p = -\mathbf{b}$
- Then any point in affine space $\boldsymbol{x}_p + ker(\mathbf{A})$ is also stationary, with same function value and same Hessian (as it is constant)



OPTIMA: RANK-DEFICIENT CASE

- All affine spaces of form $\boldsymbol{x}_p + ker(\mathbf{A})$ for diff. valid $\boldsymbol{x}_p$ are the same
- Any stationary point must be in $\boldsymbol{x}_p + ker(\mathbf{A})$
- So $\boldsymbol{x}_p + ker(\mathbf{A})$ are all the stationary points, with same curvature
- If $\mathbf{A} \succeq 0$, they are all minima
- If $\mathbf{A} \preceq 0$, they are all maxima
- If $\mathbf{A}$ is indefinite, they are all saddle points

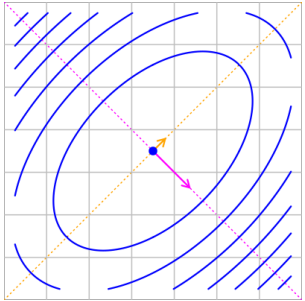


GEOMETRY OF QUADRATIC FUNCTIONS

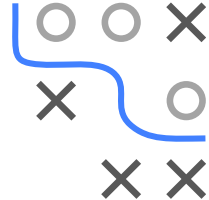
- Example: $\mathbf{A} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \Rightarrow \mathbf{H} = 2\mathbf{A} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$
- Since $\mathbf{H}$ symmetric: eigendecomposition $\mathbf{H} = \mathbf{V}\Lambda\mathbf{V}^T$

$$\mathbf{V} = \begin{pmatrix} | & | \\ v_{\max} & v_{\min} \\ | & | \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \text{ orthogonal}$$

$$\Lambda = \begin{pmatrix} \lambda_{\max} & 0 \\ 0 & \lambda_{\min} \end{pmatrix} = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$$



SPECTRUM AND CURVATURE

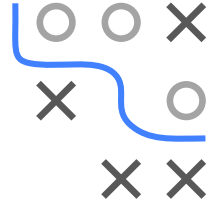


- $\boldsymbol{v}_{\max}$ direction of highest curvature, with curvature value $\lambda_{\max}$

$$\boldsymbol{v}^T\mathbf{H}\boldsymbol{v} = \boldsymbol{v}^T\mathbf{V}\Lambda\mathbf{V}^T\boldsymbol{v} = \boldsymbol{w}^T\Lambda\boldsymbol{w} = \sum_{i=1}^{d}\lambda_i w_i^2 \leq \lambda_{\max}\sum_{i=1}^{d} w_i^2 = \lambda_{\max}||\boldsymbol{w}||^2$$

- Since $||\boldsymbol{v}|| = ||\boldsymbol{x}||$ ($\mathbf{V}$ orthogonal): $\max_{||\boldsymbol{v}||=1} \boldsymbol{v}^T\mathbf{H}\boldsymbol{v} \leq \lambda_{\max}$
- For $\boldsymbol{v}_{\max}$ we obtain this upper bound: $\boldsymbol{v}_{\max}^T\mathbf{H}\boldsymbol{v}_{\max} = \mathbf{e}_1^T\Lambda\mathbf{e}_1 = \lambda_{\max}$
- Analogously, $\boldsymbol{v}_{\min}$ direction of lowest curvature, with curvature value $\lambda_{\min}$
- Contour lines of any quadratic function are ellipses

SECOND-ORDER SUFFICIENT CONDITION



- Recall: Second order condition for optimality is sufficient
- If $H(\boldsymbol{x}^*) \succ 0$ at stationary point $\boldsymbol{x}^*$, then $\boldsymbol{x}^*$ local minimum ($\prec$ for maximum)

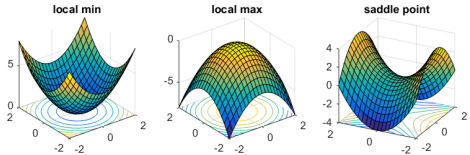
$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + \underbrace{\nabla f(\boldsymbol{x}^*)}_{=0}(\boldsymbol{x} - \boldsymbol{x}^*) + \frac{1}{2}\underbrace{(\boldsymbol{x} - \boldsymbol{x}^*)^T H(\boldsymbol{x}^*)(\boldsymbol{x} - \boldsymbol{x}^*)}_{\geq \lambda_{\min}\|\boldsymbol{x}-\boldsymbol{x}^*\|^2} + \underbrace{R_2(\boldsymbol{x}, \boldsymbol{x}^*)}_{=o(\|\boldsymbol{x}-\boldsymbol{x}^*\|^2)}$$

- Choose $\epsilon > 0$ s.t. $|R_2(\boldsymbol{x}, \boldsymbol{x}^*)| < \frac{1}{2}\lambda_{\min}\|\boldsymbol{x} - \boldsymbol{x}^*\|^2$ for $\boldsymbol{x} \neq \boldsymbol{x}^*$, $\|\boldsymbol{x} - \boldsymbol{x}^*\| < \epsilon$

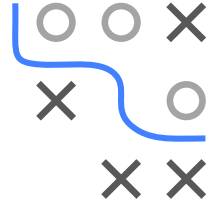
$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*) + \underbrace{\frac{1}{2}\lambda_{\min}\|\boldsymbol{x} - \boldsymbol{x}^*\|^2 + R_2(\boldsymbol{x}, \boldsymbol{x}^*)}_{>0} > f(\boldsymbol{x}^*)$$

EIGENVALUES AND SHAPE

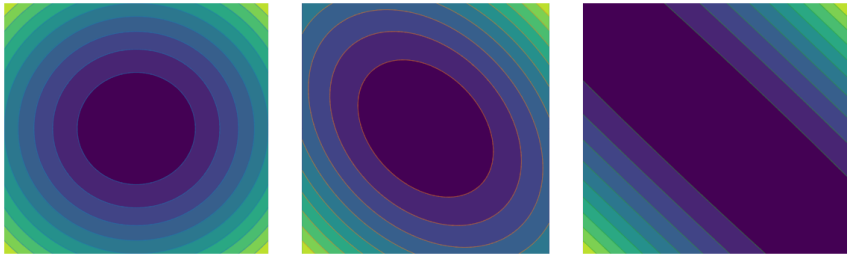
- If spectrum of $\mathbf{A}$ is known, also that of $\mathbf{H} = 2\mathbf{A}$ is known
- If all eigenvalues of $\mathbf{H} \overset{(>)}{\geq} 0$ ($\Leftrightarrow \mathbf{H} \overset{(\succ)}{\succcurlyeq} 0$):
  - q (strictly) convex
  - (Unique) global minimum
- If all eigenvalues of $\mathbf{H} \overset{(<)}{\leq} 0$ ($\Leftrightarrow \mathbf{H} \overset{(\prec)}{\preccurlyeq} 0$):
  - q (strictly) concave
  - (Unique) global maximum
- If $\mathbf{H}$ has both positive and negative eigenvalues ($\Leftrightarrow$ $\mathbf{H}$ indefinite):
  - q neither convex nor concave
  - there is a saddle point



CONDITION AND CURVATURE



- $\kappa(\mathbf{H}) = \kappa(\mathbf{A}) = |\lambda_{\text{max}}|/|\lambda_{\text{min}}|$
- High condition means
  - $|\lambda_{\text{max}}| \gg |\lambda_{\text{min}}|$
  - Curvature along $\boldsymbol{v}_{\text{max}} \gg$ along $\boldsymbol{v}_{\text{min}}$
  - Problem for algorithms like gradient descent

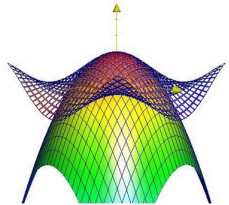


Left: Excellent condition. Middle: Good condition. Right: Bad condition.

APPROXIMATION OF SMOOTH FUNCTIONS

- Any $f \in \mathcal{C}^2$ can be locally approximated by quadratic function (second order Taylor)

$$f(\boldsymbol{x}) \approx f(\tilde{\boldsymbol{x}}) + \nabla f(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}}) + \frac{1}{2}(\boldsymbol{x} - \tilde{\boldsymbol{x}})^T \nabla^2 f(\tilde{\boldsymbol{x}})(\boldsymbol{x} - \tilde{\boldsymbol{x}})$$



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f and second order approximation: dark vs bright grid.

- $\implies$ Hessians provide information about **local** geometry of a function